www.tntextbooks.in

$$4\ H_2O \longrightarrow 4\ H^+ + 4\ OH^-$$
$$4\ OH^- \longrightarrow 2\ H_2O + O_2 + 4\ e^-$$
$$2H_2O \longrightarrow 4\ H^+ + O_2 + 4\ e^-$$

13.10.2 Electron Transport Chain of Chloroplast

PQ acts as shuttle between PS II and Cytochrome b_6- f complex and PC connects

- Cytochrome b_6-f and PS I complex.
- *ATPase complex or Coupling factor:* It is found in the surface of thylakoid membrane. This complex is made up

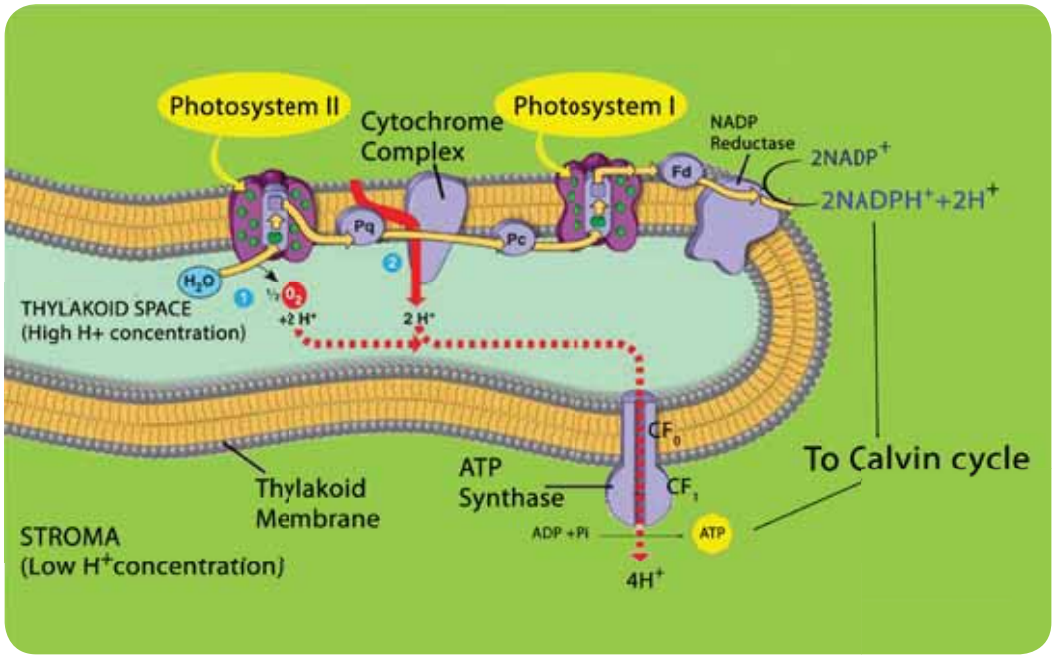


Figure 13.14: Electron Transport Chain in Chloroplast

Electron transport chain in each photosystem involves four complexes:

- *Core Complex (CC):* CC I in PS I the reaction centre is P700, CC II in PS II the reaction centre is P680
- Light Harvesting Complex or Antenna complex (LHC):
- Two types: LHC I in PS I and LHC II in PS II.
- Cytochrome b6 f complex: It is the non-pigmented protein complex connecting PS I and PS II. Plastoquinone (PQ) and Plastocyanin (PC) are intermediate complexes acting as mobile or shuttle electron carriers of Electron Transport Chain.

of CF_1 and CF_0 factors. This complex utilizes energy from ETC and converts ADP and inorganic phosphate (P_i) into ATP (Figure 13.14).

13.11 Photophosphorylation

Phosphorylation taking place during respiration is called as **oxidative phosphorylation** and ATP produced by the breakdown of substrate is known as **substrate level phosphorylation**. In this topic, we are going to learn about phosphorylation taking place in chloroplast with the help of light. During the movement of electrons through carrier molecules ATP and NADPH + H^+ are